Picture Perfect

Fermat's Last Theorem and
Beal's Conjecture

$\overset{\theta}{\delta}\Delta^{\Theta}_{\beta}$

Dempster Welling

Thanksgivings 2021

1 Representing Perfects

$$\beta^{\theta}$$

$$\sum_{r=1}^{\beta-1} S_r + 1$$

$$\sum_{l=1}^{\beta-1} S_l + 1$$

2 Computational Algorithm

$$\boxed{H_\Delta = \begin{cases} 0 & \theta \text{ zero,} \\ \Delta(\theta, \beta, \delta) & \theta \text{ whole.} \end{cases}}$$

$$\boxed{\Delta(\theta, \beta, \delta) = (\beta + \delta)\Delta(\theta - 1, \beta, \delta) + \delta \sum_{r=1}^{\beta-1} \Delta(\theta - 1, r, \delta)}$$

$$\boxed{P^\theta - P^{(\theta-2)} = P^{(\theta-2)}(P^2 - 1)}$$

$$\boxed{C_\theta = P^\theta - P^{(\theta-2)}}$$

$$\boxed{C_\theta \mod 6 = 0}$$

$$\boxed{C_\theta \quad has \quad six \quad identical \quad parts}$$

$$\boxed{S_\theta = \Delta(\theta, \beta, \delta) - \Delta((\theta - 2), \beta, \delta)}$$

3 Construction

The Δ^{whole} perfect is constructed as layers of concentric rings and an apex. The ring weights(ϵ_{ring}) are arranged in a bi-lateral symmetric representation. Two forms are possible. Odd θ or Even θ

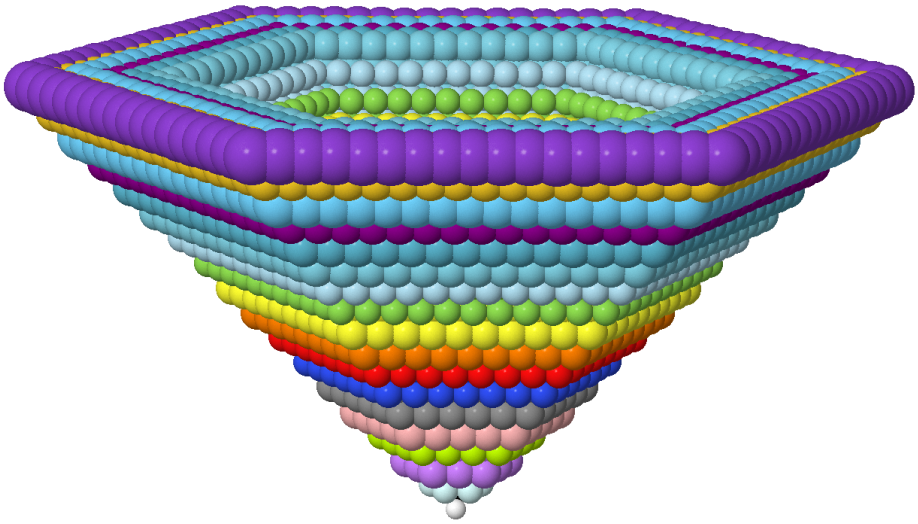
- $\Delta^{odd} = \sum_r^{odd} C_\beta^r + (\beta^1)_{def} + \alpha \quad 6s - axial(linear)$
- $\Delta^{even} = \sum_r^{even} C_\beta^r + (\beta^2)_{def} + \alpha \quad 2s - bilateral(triangularsquare)$

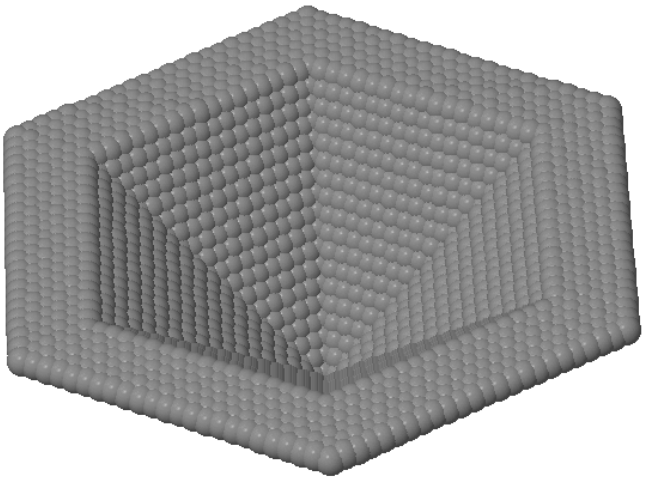
3

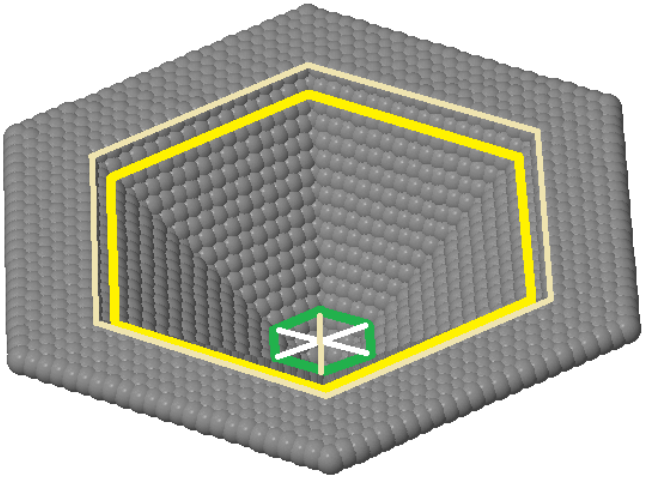
$$H_\Delta = \begin{cases} 0 & \theta \text{ zero,} \\ \Delta(\theta, \beta, \delta) & \theta \text{ whole.} \end{cases}$$

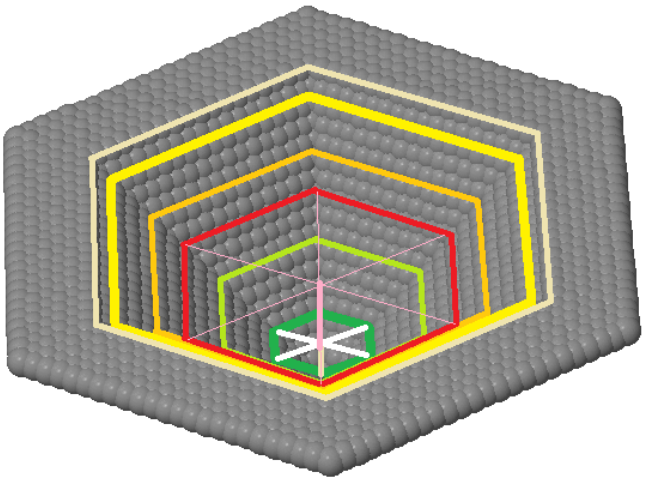
ϵ_0	ϵ_1	ϵ_2	ϵ_3	ϵ_4	ϵ_5	ϵ_6	ϵ_7	ϵ_8
α_0								
ω_0	ω_1							
ω_0	ω_1	ω_2						
ω_0	ω_1	ω_2	ω_3					
ω_0	ω_1	ω_2	ω_3	ω_4				
ω_0	ω_1	ω_2	ω_3	ω_4	ω_5			
ω_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6		
ω_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	
ω_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8

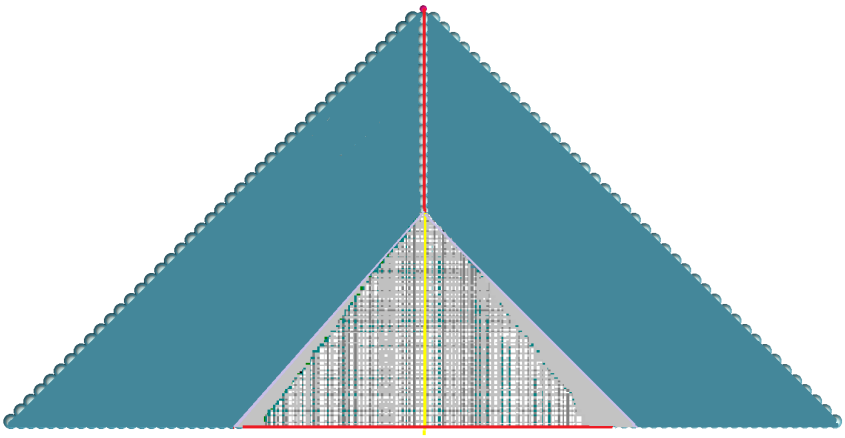
ϵ_8	ϵ_7	ϵ_6	ϵ_5	ϵ_4	ϵ_3	ϵ_2	ϵ_1	ϵ_0	ϵ_1	ϵ_2	ϵ_3	ϵ_4	ϵ_5	ϵ_6	ϵ_7	ϵ_8
								α_0								
							ω_1	ω_0	ω_1							
						ω_2	ω_1	ω_0	ω_1	ω_2						
					ω_3	ω_2	ω_1	ω_0	ω_1	ω_2	ω_3					
				ω_4	ω_3	ω_2	ω_1	ω_0	ω_1	ω_2	ω_3	ω_4				
			ω_5	ω_4	ω_3	ω_2	ω_1	ω_0	ω_1	ω_2	ω_3	ω_4	ω_5			
		ω_6	ω_5	ω_4	ω_3	ω_2	ω_1	ω_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6		
	ω_7	ω_6	ω_5	ω_4	ω_3	ω_2	ω_1	ω_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	
ω_8	ω_7	ω_6	ω_5	ω_4	ω_3	ω_2	ω_1	ω_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8











δ 1	0	1	2	3	4	5	6	7	8
1	1	1	1	1	1	1	1	1	1
2	1	3	5	7	9	11	13	15	17
3	1	7	19	37	61	91	127	169	217
4	1	15	65	175	369	671	1105	1695	2465
5	1	31	211	781	2101	4651	9031	15961	26281
6	1	63	665	3367	11529	31031	70993	144495	269297
7	1	127	2059	14197	61741	201811	543607	1273609	2685817
8	1	255	6305	58975	325089	1288991	4085185	11012415	26269505

δ 2	0	1	2	3	4	5	6	7	8
1	2	2	2	2	2	2	2	2	2
2	4	8	12	16	20	24	28	32	36
3	8	26	56	98	152	218	296	386	488
4	16	80	240	544	1040	1776	2800	4160	5904
5	32	242	992	2882	6752	13682	24992	42242	67232
6	64	728	4032	14896	42560	102024	215488	413792	737856
7	128	2186	16256	75938	263552	745418	1817216	3959426	7902848
8	256	6560	65280	384064	1614080	5374176	15097600	37281920	83222784

δ 3	0	1	2	3	4	5	6	7	8
1	3	3	3	3	3	3	3	3	3
2	9	15	21	27	33	39	45	51	57
3	27	63	117	189	279	387	513	657	819
4	81	255	609	1215	2145	3471	5265	7599	10545
5	243	1023	3093	7533	15783	29643	51273	83193	128283
6	729	4095	15561	45927	113553	246519	484785	882351	1509417
7	2187	16383	77997	277749	807159	2019027	4503033	9176457	17390019
8	6561	65535	390369	1673055	5699265	16386591	41367105	94235199	197581665

δ 4	0	1	2	3	4	5	6	7	8
1	4	4	4	4	4	4	4	4	4
2	16	24	32	40	48	56	64	72	80
3	64	124	208	316	448	604	784	988	1216
4	256	624	1280	2320	3840	5936	8704	12240	16640
5	1024	3124	7744	16564	31744	55924	92224	144244	216064
6	4096	15624	46592	116920	258048	515816	953344	1653912	2723840
7	16384	78124	279808	821356	2080768	4704844	9720064	18663628	33734656
8	65536	390624	1679360	5758240	16711680	42656096	98320384	208594080	413204480

δ 5	0	1	2	3	4	5	6	7	8
1	5	5	5	5	5	5	5	5	5
2	25	35	45	55	65	75	85	95	105
3	125	215	335	485	665	875	1115	1385	1685
4	625	1295	2385	4015	6305	9375	13345	18335	24465
5	3125	7775	16775	32525	58025	96875	153275	232025	338525
6	15625	46655	117585	261415	527345	984375	1724905	2868335	4564665
7	78125	279935	823415	2094965	4766585	9921875	19207235	35008265	60651365
8	390625	1679615	5764545	16770655	42981185	99609375	212679265	424216895	798953505

δ 6	0	1	2	3	4	5	6	7	8
1	6	6	6	6	6	6	6	6	6
2	36	48	60	72	84	96	108	120	132
3	216	342	504	702	936	1206	1512	1854	2232
4	1296	2400	4080	6480	9744	14016	19440	26160	34320
5	7776	16806	32736	58806	98976	157926	241056	354486	505056
6	46656	117648	262080	530712	995904	1755936	2939328	4709160	7267392
7	279936	823542	2097024	4780782	9983616	19409046	35551872	61924974	103316352
8	1679616	5764800	16776960	43040160	99934464	213968256	428302080	809965920	1459011840

δ 7	0	1	2	3	4	5	6	7	8
1	7	7	7	7	7	7	7	7	7
2	49	63	77	91	105	119	133	147	161
3	343	511	721	973	1267	1603	1981	2401	2863
4	2401	4095	6545	9919	14385	20111	27265	36015	46529
5	16807	32767	59017	99757	160027	245707	363517	521017	726607
6	117649	262143	531377	999271	1767465	2970359	4780153	7411887	11128481
7	823543	2097151	4782841	9997813	19470787	35753683	62468581	104589961	168762223
8	5764801	16777215	43046465	99993439	214293345	429591071	814051105	1470024255	2546113409

δ 8	0	1	2	3	4	5	6	7	8
1	8	8	8	8	8	8	8	8	8
2	64	80	96	112	128	144	160	176	192
3	512	728	992	1304	1664	2072	2528	3032	3584
4	4096	6560	9984	14560	20480	27936	37120	48224	61440
5	32768	59048	99968	160808	247808	368168	530048	742568	1015808
6	262144	531440	999936	1770832	2981888	4811184	7482880	11272976	16515072
7	2097152	4782968	9999872	19484984	35815424	62670392	105133568	170035832	266338304
8	16777216	43046720	99999744	214352320	429916160	815340096	1474109440	2557125824	4278190080

Figure 1: Flawless First Article Relic : Diagram

${}_{\theta}\epsilon_1$	2	3	4	5	6	7	8	9
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	4	28	92	216	420	724	1148	1712
6	8	92	432	1328	3200	6588	12152	20672
7	16	292	1928	7704	22920	56236	120512	233648
8	32	908	8328	42992	157400	459372	1142288	2522048

${}_{\theta}\epsilon_2$	2	3	4	5	6	7	8	9
3	3	5	7	9	11	13	15	17
4	10	26	50	82	122	170	226	290
5	32	120	308	636	1144	1872	2860	4148
6	100	524	1760	4528	9788	18740	32824	53720
7	308	2220	9632	30624	79156	176748	354160	653072
8	940	9236	51320	200392	616772	1601660	3664336	7602800

${}_{\theta}\epsilon_3$	2	3	4	5	6	7	8	9
3	6	9	12	15	18	21	24	27
4	28	58	100	154	220	298	388	490
5	124	336	728	1360	2292	3584	5296	7488
6	532	1852	4960	11116	21940	39412	65872	104020
7	2236	9924	32552	86860	199668	410396	773584	1361412
8	9268	52228	208720	659764	1759060	4123708	8745088	17119900

${}_{\theta}\epsilon_4$	2	3	4	5	6	7	8	9
3	10	14	18	22	26	30	34	38
4	60	108	172	252	348	460	588	732
5	340	756	1452	2508	4004	6020	8636	11932
6	1860	5052	11548	23268	42612	72460	116172	177588
7	9940	32844	88788	207372	433316	829820	1481924	2499868
8	52260	209628	668092	1802052	4281108	9204460	18262188	33923412

${}_{\theta}\epsilon_5$	2	3	4	5	6	7	8	9
3	15	20	25	30	35	40	45	50
4	110	180	270	380	510	660	830	1020
5	760	1480	2600	4220	6440	9360	13080	17700
6	5060	11640	23700	43940	75660	122760	189740	281700
7	32860	89080	209300	441020	852740	1538160	2620380	4256100
8	209660	669000	1810380	4324100	9361860	18721560	35065700	62173380

${}_{\theta}\epsilon_6$	2	3	4	5	6	7	8	9
3	21	27	33	39	45	51	57	63
4	182	278	398	542	710	902	1118	1358
5	1484	2628	4312	6656	9780	13804	18848	25032
6	11648	23792	44372	76988	125960	196328	293852	425012
7	89096	209592	442948	860444	1561080	2676616	4376612	6873468
8	669032	1811288	4332428	9404852	18878960	35525072	63315668	107747948

${}_{\theta}\epsilon_7$	2	3	4	5	6	7	8	9
3	28	35	42	49	56	63	70	77
4	280	406	560	742	952	1190	1456	1750
5	2632	4340	6748	9996	14224	19572	26180	34188
6	23800	44464	77420	127288	199528	300440	437164	617680
7	209608	443240	862372	1568784	2699536	4432848	6993980	10661112
8	1811320	4333336	9413180	18921952	35682472	63775040	108890236	178728760

${}_{\theta}\epsilon_8$	2	3	4	5	6	7	8	9
3	36	44	52	60	68	76	84	92
4	408	568	760	984	1240	1528	1848	2200
5	4344	6776	10088	14440	19992	26904	35336	45448
6	44472	77512	127720	200856	303640	443752	629832	871480
7	443256	862664	1570712	2707240	4455768	7050216	10781624	16004872
8	4333368	9414088	18930280	35725464	63932440	109349608	179871048	285970360

Figure 2: Flawless First Article Relic : Diagram

								L	C	R						
							L	L	C	R	R					
							L	W	C	W	R					
						L	L	L	C	R	R	R				
						L	L	W	C	W	R	R				
						L	W	W	C	W	W	R				
					L	L	L	L	C	R	R	R	R			
					L	L	L	W	C	W	R	R	R			
					L	L	W	W	C	W	W	R	R			
					L	W	W	W	C	W	W	W	R			
				L	L	L	L	L	C	R	R	R	R	R		
				L	L	L	L	W	C	W	R	R	R	R		
				L	L	L	W	W	C	W	W	R	R	R		
				L	L	W	W	W	C	W	W	W	R	R		
				L	W	W	W	W	C	W	W	W	W	R		
			L	L	L	L	L	L	C	R	R	R	R	R	R	
			L	L	L	L	L	W	C	W	R	R	R	R	R	
			L	L	L	L	W	W	C	W	W	R	R	R	R	
			L	L	L	W	W	W	C	W	W	W	R	R	R	
			L	L	W	W	W	W	C	W	W	W	W	R	R	
			L	W	W	W	W	W	C	W	W	W	W	W	R	
		L	L	L	L	L	L	L	C	R	R	R	R	R	R	R
		L	L	L	L	L	L	W	C	W	R	R	R	R	R	R
		L	L	L	L	L	W	W	C	W	W	R	R	R	R	R
		L	L	L	L	W	W	W	C	W	W	W	R	R	R	R
		L	L	L	W	W	W	W	C	W	W	W	W	R	R	R
		L	W	W	W	W	W	W	C	W	W	W	W	W	W	R
L	L	L	L	L	L	L	L	L	C	R	R	R	R	R	R	R
L	L	L	L	L	L	L	W	W	C	W	R	R	R	R	R	R
L	L	L	L	L	L	W	W	W	C	W	W	R	R	R	R	R
L	L	L	L	L	W	W	W	W	C	W	W	W	R	R	R	R
L	L	L	L	W	W	W	W	W	C	W	W	W	W	R	R	R
L	L	L	W	W	W	W	W	W	C	W	W	W	W	W	R	R
L	L	W	W	W	W	W	W	W	C	W	W	W	W	W	W	R
L	W	W	W	W	W	W	W	W	C	W	W	W	W	W	W	R

Figure 3: Flawless First Article Relic : Diagram

								L	Δ	R						
							L	L	Δ	R	R					
							L	W	α	W	R					
						L	L	L	Δ	R	R	R				
						L	L	W	${}_lC$	W	R	R				
						L	W	W	α	W	W	R				
					L	L	L	L	Δ	R	R	R	R			
					L	L	L	W	${}_lC$	W	R	R	R			
					L	L	W	W	C_r	W	W	R	R			
					L	W	W	W	α	W	W	W	R			
				L	L	L	L	L	Δ	R	R	R	R	R		
				L	L	L	L	W	${}_lC$	W	R	R	R	R		
				L	L	L	W	W	C_r	W	W	R	R	R		
				L	L	W	W	W	${}_lC$	W	W	W	R	R		
				L	W	W	W	W	α	W	W	W	W	R		
			L	L	L	L	L	L	Δ	R	R	R	R	R	R	
			L	L	L	L	L	W	${}_lC$	W	R	R	R	R	R	
			L	L	L	L	W	W	C_r	W	W	R	R	R	R	
			L	L	L	W	W	W	${}_lC$	W	W	W	R	R	R	
			L	L	W	W	W	W	C_r	W	W	W	W	R	R	
			L	W	W	W	W	W	α	W	W	W	W	W	R	
		L	L	L	L	L	L	L	Δ	R	R	R	R	R	R	R
		L	L	L	L	L	L	W	${}_lC$	W	R	R	R	R	R	R
		L	L	L	L	L	W	W	C_r	W	W	R	R	R	R	R
		L	L	L	L	W	W	W	${}_lC$	W	W	W	R	R	R	R
		L	L	L	W	W	W	W	C_r	W	W	W	W	R	R	R
		L	L	W	W	W	W	W	${}_lC$	W	W	W	W	W	R	R
		L	W	W	W	W	W	W	α	W	W	W	W	W	W	R
	L	L	L	L	L	L	L	L	Δ	R	R	R	R	R	R	R
	L	L	L	L	L	L	L	W	${}_lC$	W	R	R	R	R	R	R
	L	L	L	L	L	L	W	W	C_r	W	W	R	R	R	R	R
	L	L	L	L	L	W	W	W	${}_lC$	W	W	W	R	R	R	R
	L	L	L	L	W	W	W	W	C_r	W	W	W	W	R	R	R
	L	L	L	W	W	W	W	W	${}_lC$	W	W	W	W	W	R	R
	L	L	W	W	W	W	W	W	α	W	W	W	W	W	W	R
L	L	L	L	L	L	L	L	L	Δ	R	R	R	R	R	R	R
L	L	L	L	L	L	L	L	W	${}_lC$	W	R	R	R	R	R	R
L	L	L	L	L	L	L	W	W	C_r	W	W	R	R	R	R	R
L	L	L	L	L	L	W	W	W	${}_lC$	W	W	W	R	R	R	R
L	L	L	L	L	W	W	W	W	C_r	W	W	W	W	R	R	R
L	L	L	L	W	W	W	W	W	${}_lC$	W	W	W	W	W	R	R
L	L	L	W	W	W	W	W	W	C_r	W	W	W	W	W	R	R
L	L	W	W	W	W	W	W	W	α	W	W	W	W	W	W	R

Figure 4: Flawless First Article Relic : Diagram

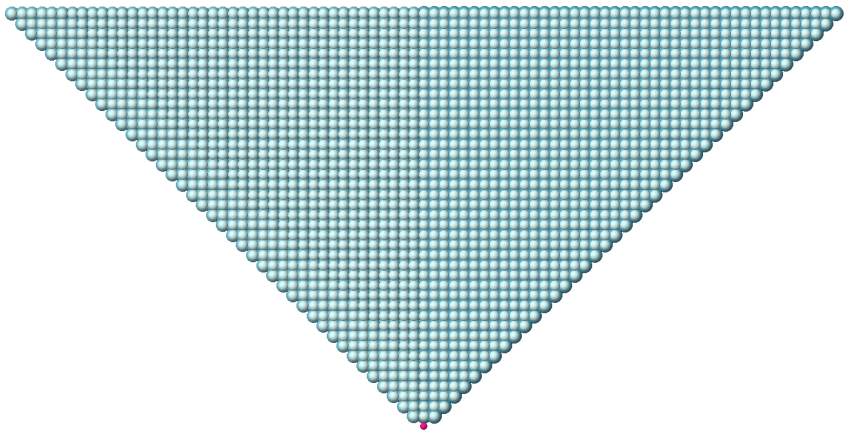


Figure 5: Bilateral $Perfect^{2nd}$

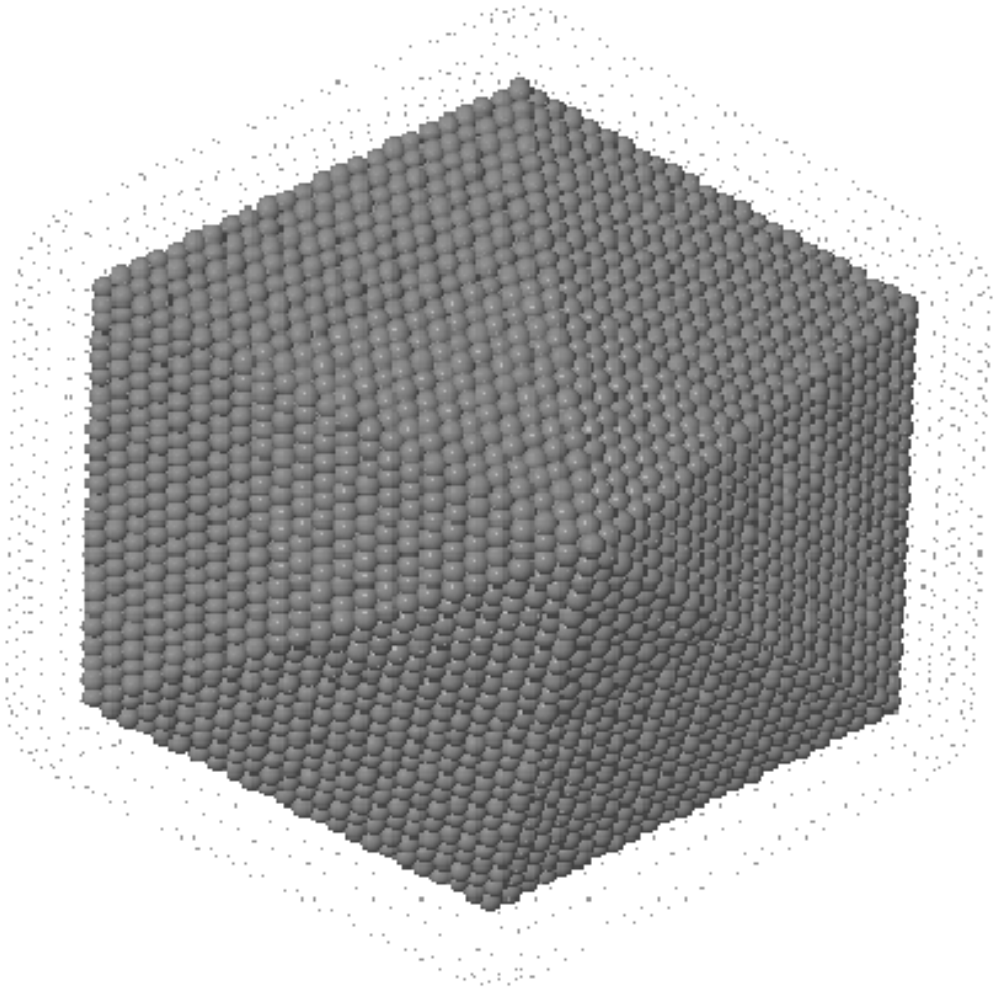


Figure 6: Hexagonal $Perfect^{3rd}$

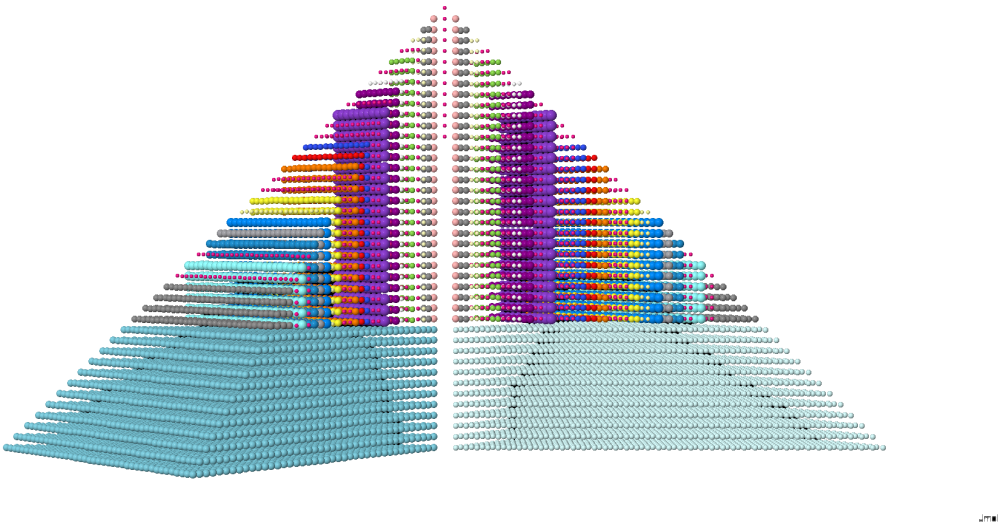


Figure 7: Hexagonal $Perfect^{Nth}$ partial construction

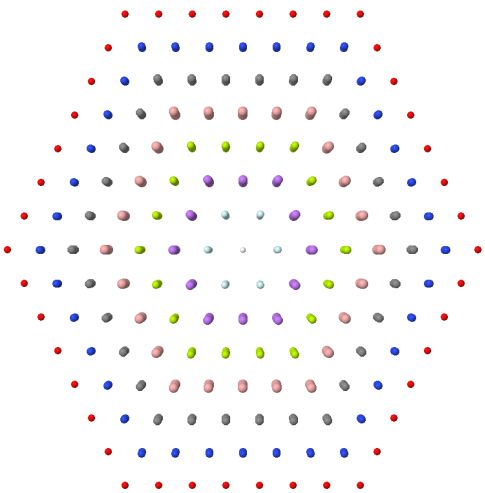


Figure 8: Hexagonal $Perfect^{Nth}$ top view

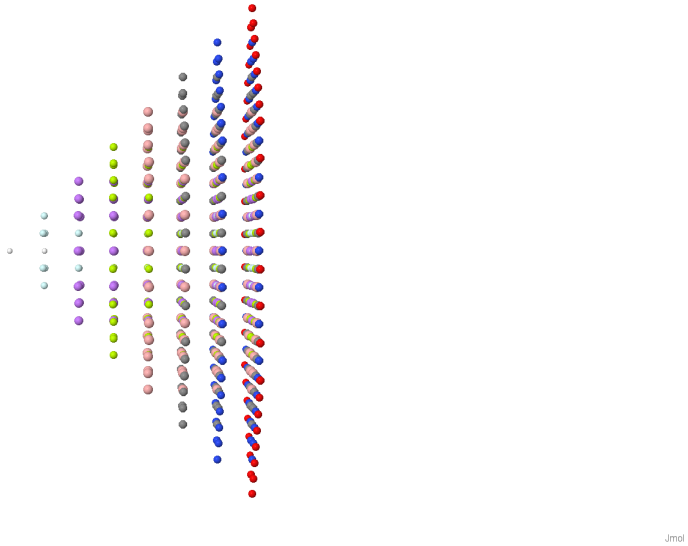


Figure 9: Hexagonal $Perfect^{Nth}$ side view 1

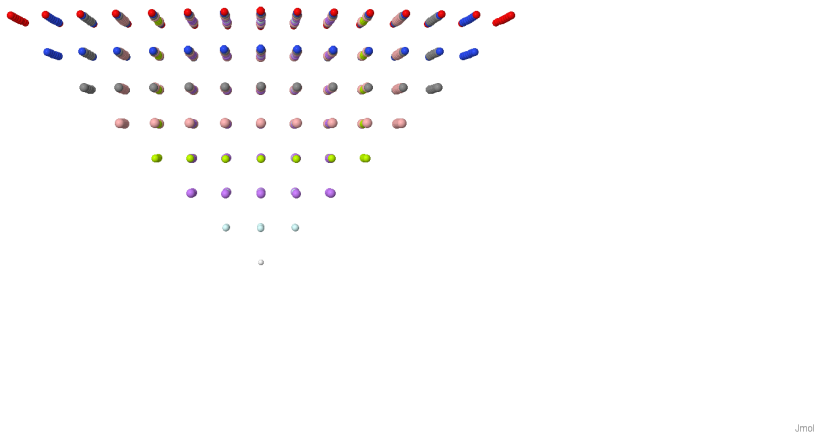


Figure 10: Hexagonal $Perfect^{Nth}$ side view 2